

Year 10 Science (Adjusted)
Semester 1 Examination 2020 - Revision

Earth and Space Sciences

1. Complete the following table.

TERM	DEFINITION
light year	
constellation	
galaxy	
nebula	
supernova	
red giant	
black hole	
fusion	

2. Light travels at **300,000 km/second**. A star is 97.6 light years away. We can only travel at about **15 - 20 km/second** with our best rockets. No calculations are necessary.

(a) With our current technology, are we likely to be able to travel to this star in the near future? Give a few reasons for your answer.

YES / NO (Circle your answer.)

Reasons: •

•

•

- (b) Suggest **one way** we might be able to overcome the time factor of travelling to other galaxies, given it could be hundreds of years.

3. (a) What is the major element found in the Sun? _____
- (b) What happens to this element so that the Sun produces its energy?
(HINT: Think about the fusion process.)
- (c) The Sun loses 5 million tonnes of mass per second. According to Einstein's equation ($E = mc^2$), what does this mass become?

4. How is the surface temperature of a star related to its colour?

TEMPERATURE	COLOUR
	blue
	red
	yellow

5. (a) About how old is our Sun (in millions of years)? _____ million years
- (b) How much longer will it "live"? _____ million years

6. Use the following words to complete the following flow diagrams.

white dwarf black dwarf nova supernova red giant

black hole red supergiant neutron star

(a) Draw a flow diagram (with arrows) to show the stages of our Sun's remaining life as it runs out of fuel.

Sun → → → →

(b) Do similar flow diagrams for the following types of stars.

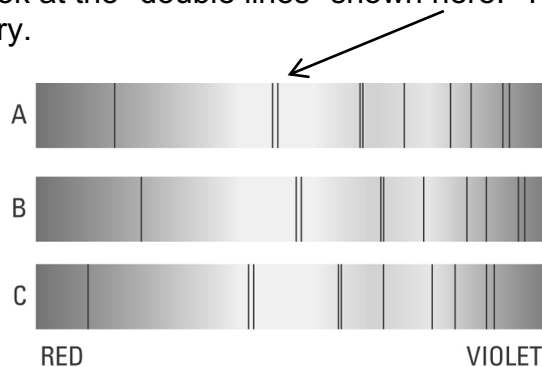
Huge star $\rightarrow$ $\rightarrow$ $\rightarrow$

Giant star → → →

7. Give two characteristics of a **black hole**.

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8. The diagram below shows the spectra (black lines) seen on the light coming from distant galaxies. Look at the "double lines" shown here. This shows the lines as seen in our laboratory.



Look at B and C above and where the double lines have moved.

(a) Circle the lines that have been "red-shifted".
(1)

(b) Does "red-shift" mean the galaxy is moving towards us? Circle your answer.

YES NO

(1)

9. Below is a list of the four spheres on Earth. Give a brief description of each.
i.e. What do you find in each one?

Biosphere:

Atmosphere:

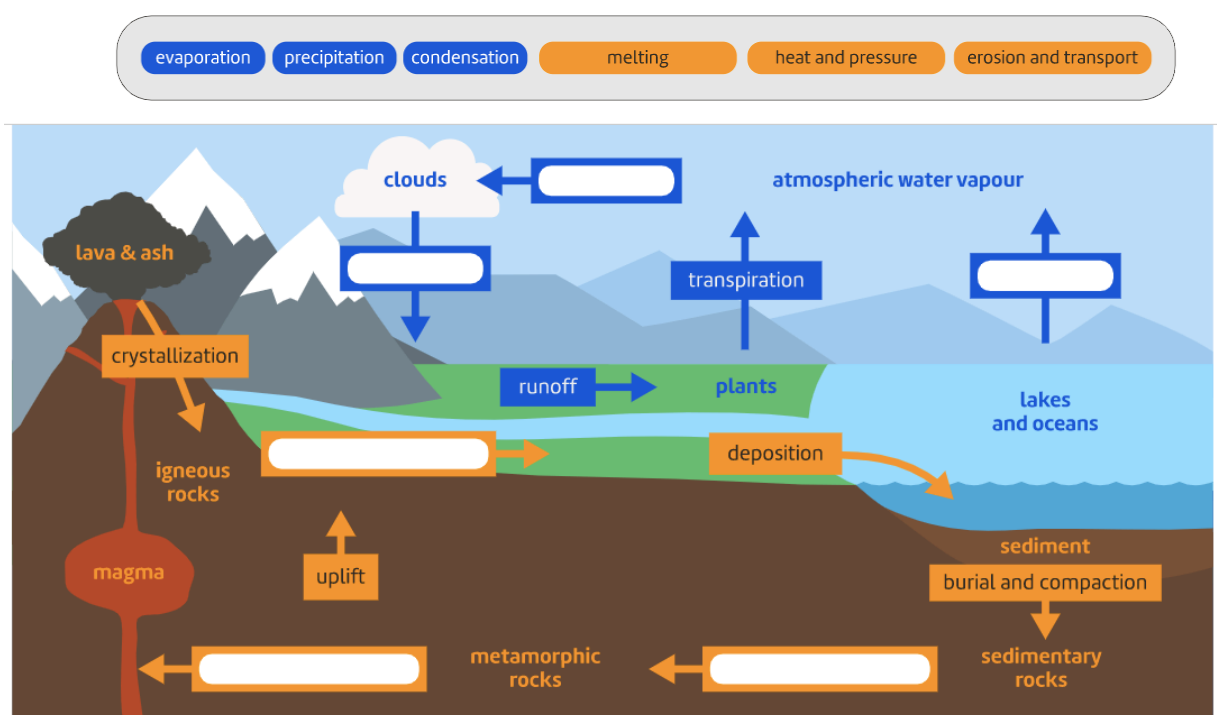
Hydrosphere:

Lithosphere:

10. Describe two consequences of **global warming**.

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-

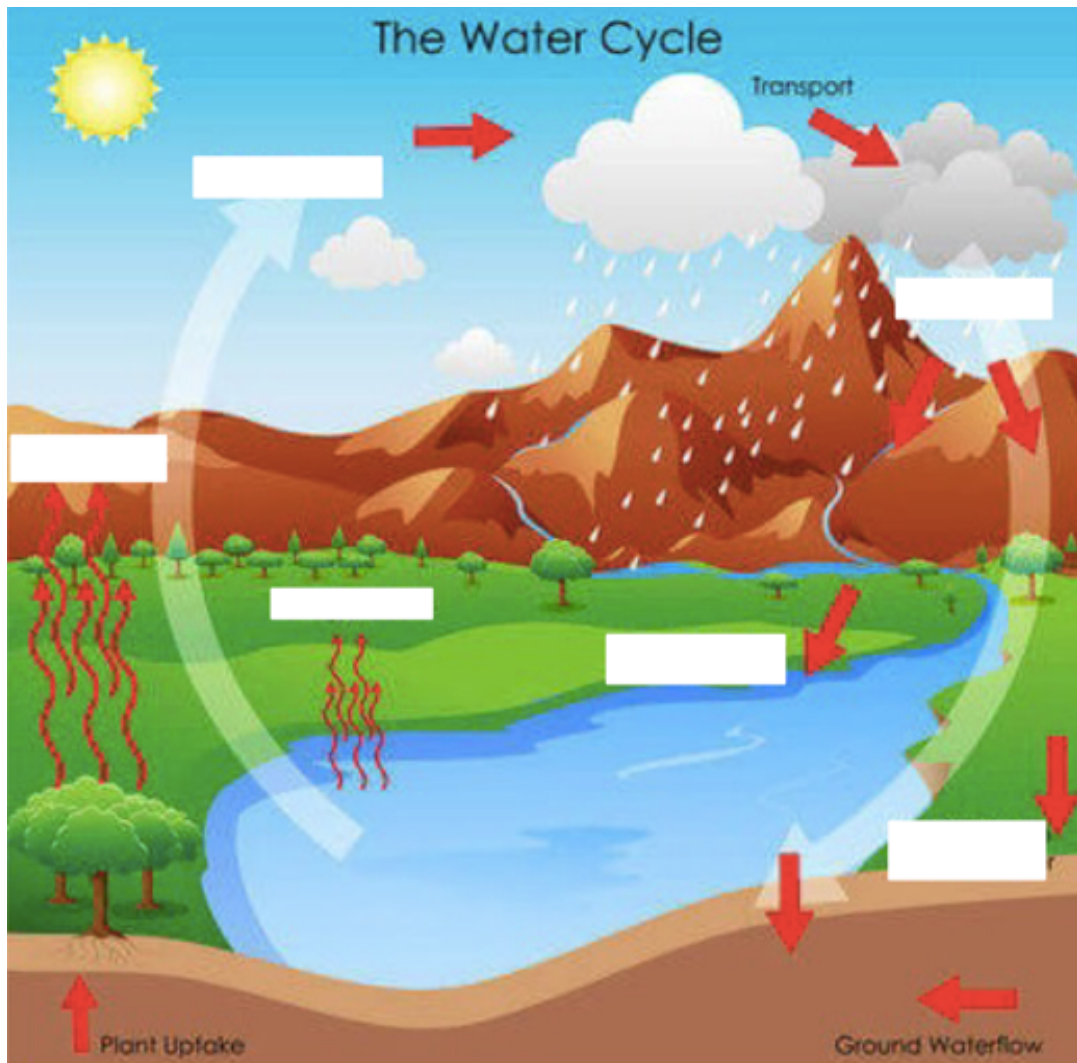
11. The following diagram shows the rock cycle and the water cycle. Complete the diagram with appropriate labels.



12. (a) Label the diagram below showing the water cycle. Us the following words.

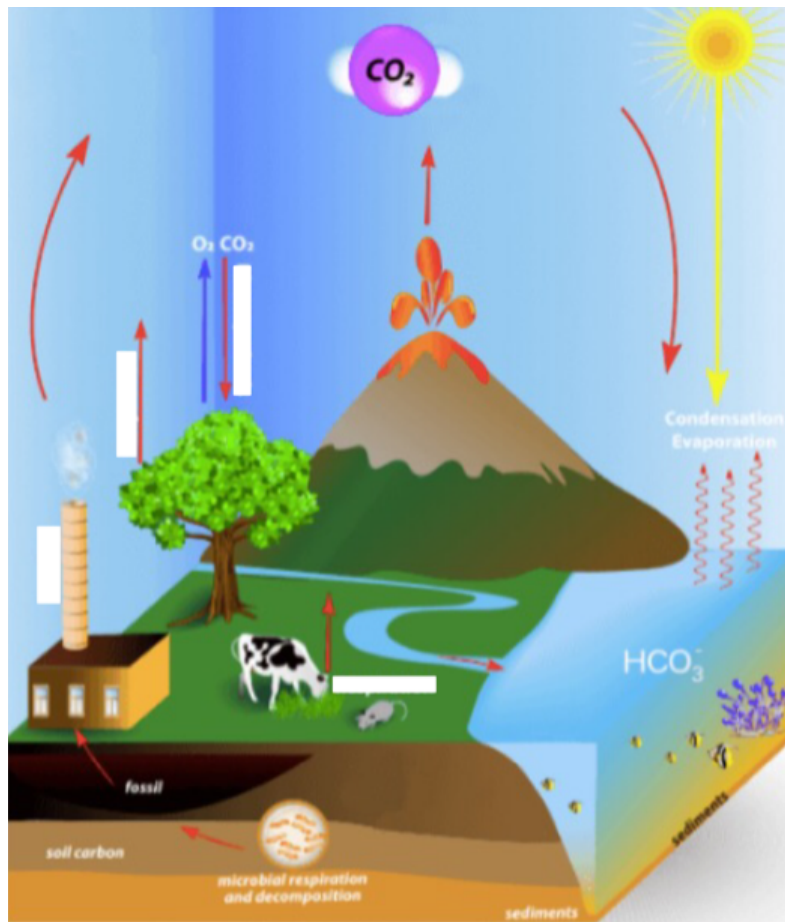
evaporation **infiltration** **precipitation** **runoff**

transpiration **condensation**



13. The following diagram shows the carbon cycle. Label the processes missing from the diagram using the words below.

combustion **photosynthesis** **respiration**



14. Use the words in the following to complete the sentences.

water rocks lithosphere glaciers hydrosphere

The Earth's [] is made up of all the [] on our planet. This includes all of the salt water oceans, fresh water lakes, [] and underground deposits.

The [] includes all [] and geological structures on which life exists, such as soil, rocks and mountains.

gases sun biosphere stable atmosphere unstable weather

The [] is made up of all areas of the earth and its atmosphere that contain life.

The [] contains all the [] above Earth. Parts of the atmosphere contain the [] and greenhouse gases that keep Earth's temperatures [].

ocean loses evaporates gains condenses

Water in the [] is heated by the sun. It [] and forms water vapour.

Clouds form when water vapour condenses. This happens because the water vapour [] heat energy as it rises.

quickly ocean evaporate rain puddle

When water precipitates as [] it either soaks into the ground or falls into a lake, river or ocean. If it falls into an [] it could take a very long time to [], but if it falls into a [] it could evaporate very [].

flows Runoff Infiltration

[] occurs when water is absorbed by the soil.

[] occurs when water [] across the ground into a lake, river, or ocean.

leaves transpiration flowers water vapour steam infiltration roots

Water is returned to the water cycle after it is absorbed by soil into the [] of a plant, it then travels through the plant and escapes through the [] of the plant as [].

This is called [].

Human Biology

1. (a) What are the male and female **gametes** in animals called?

(b) What are the **genotypes** (actual genes) for a male and a female?

2. DNA is thought to be a twisted ladder structure called a double helix. There are a series of bases located between the two strands of the helix.

(a) Name the four bases.

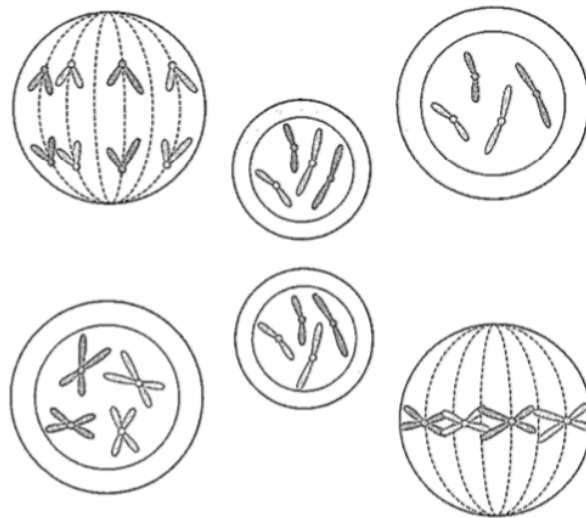
G A T C

(b) Which bases are paired together?

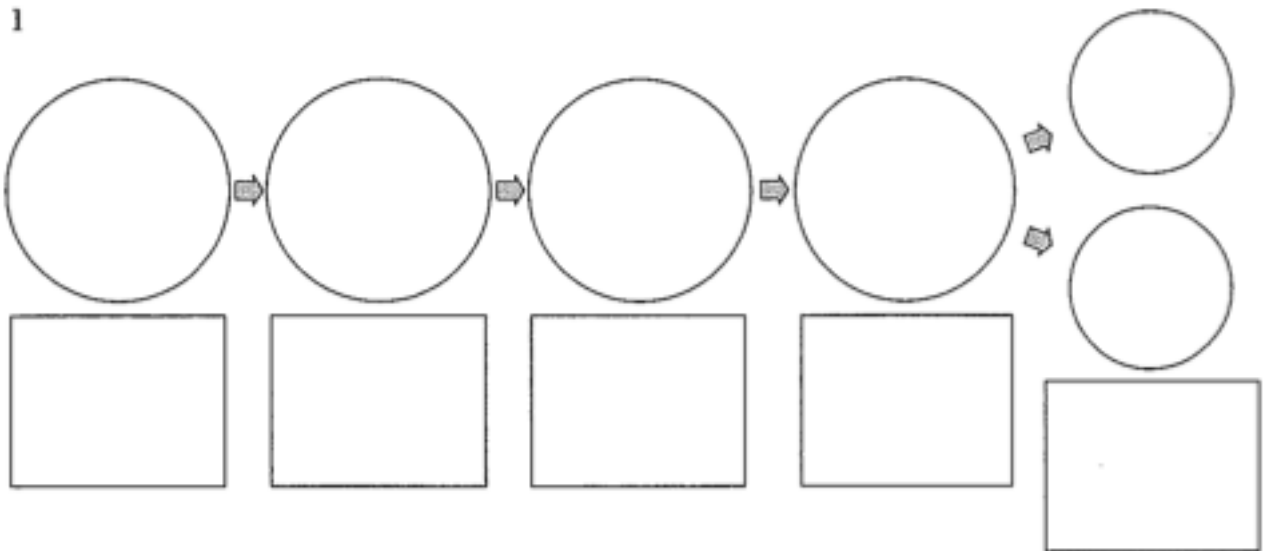
3. Define the following terms.

allele	
pure-bred	
hybrid	
homozygous	
heterozygous	
genotype	
phenotype	
sex-linked	
dominant	
recessive	
co-dominant	
incomplete dominance	
haploid	
gene	

4. Below is a diagram showing the **mitosis** cell division process. The stages are in the wrong order. Try to put them in the correct order in the blank diagram.

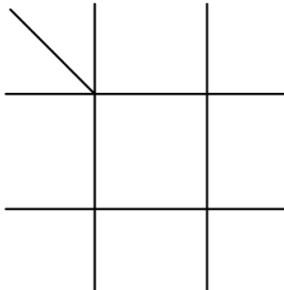


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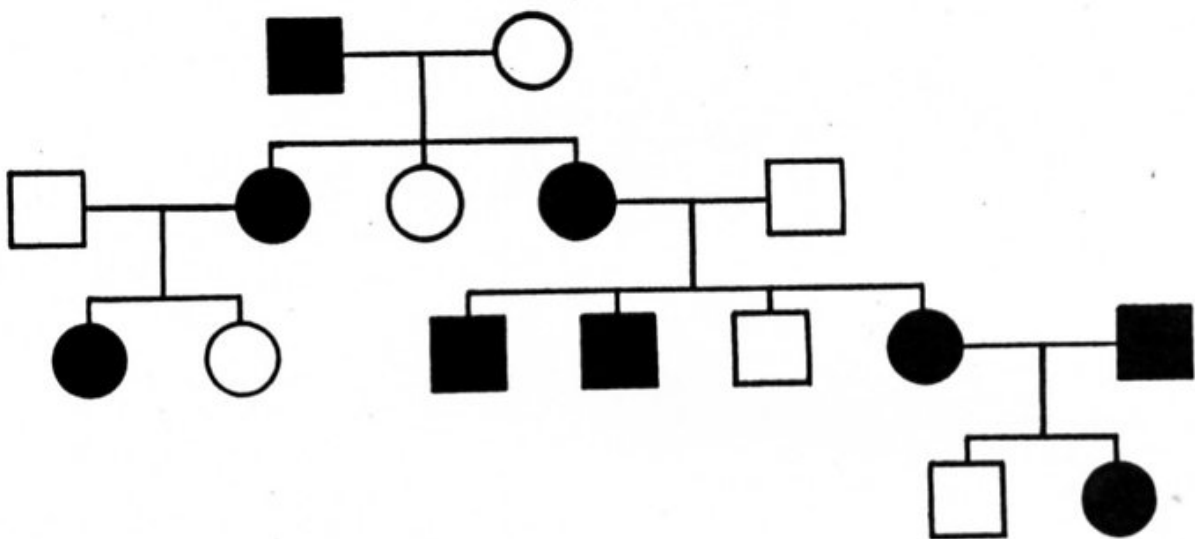
5. Polled cattle do not have horns. This is a **dominant** phenotype arising from the action of a gene **P**. Horned cattle are therefore **pp** hybrids.

What are the offspring of a hybrid polled bull (Pp) crossed with a pure-bred horned cow (pp)? **Show full working.**



6. The following pedigree shows the occurrence of no ear lobes in a family. It is a **dominant** trait.

i.e. E = no ear lobes e = ear lobes



(a) How many males have no ear lobes? _____

(b) How many females have lobes? _____

(c) What is the genotype for an individual with lobes? _____

7. (a) How many chromosomes are found in a human skin cell (a normal cell)?

(b) How many chromosomes are found in the reproductive cells?

(c) Name the process used to make reproductive cells? m_____

8. (a) Who determines the sex of a child - the mother or father? Explain why.

(b) A family has had five boys born into it, and the mother is pregnant with a sixth child. What is the chance that the child will be a girl?
